

Reed–Solomon Deep Holes and Reconstruction: Results Established through 23 July 2026

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23 July 2026

Abstract

This note summarizes the established results of a programme on projective Reed–Solomon deep holes, determinant invariants, and reconstruction from deepest-syndrome data. There are three principal theorem packages: a complete all-field redundancy-three reconstruction theory; a complete projective-semilinear classification of the deep holes of $\text{PRS}(q-4)$, of redundancy five; and all-field/high-field classifications in redundancies six and seven. The redundancy-seven theorem is complete for $q \geq 37$, while an orbit-reduced calibration below 37 remains open. We state the mathematical results, indicate the proof methods, and distinguish theorem proofs from bounded computer-assisted classifications.

1 Coding-theoretic framework

For a projective Reed–Solomon code of length $q+1$ and redundancy r , the parity-check columns form the rational normal curve

$$\nu_{r-1}(\mathbf{P}^1(\mathbf{F}_q)) \subset \mathbf{P}^{r-1}(\mathbf{F}_q).$$

A projective syndrome is deep precisely when it lies in no span of $r-2$ distinct rational points of this curve. Equivalently, the corresponding Hankel-kernel linear system of binary forms has no totally split squarefree member of degree $r-1$.

For redundancy three, let A be a six-arc in $\mathbf{P}^2(\mathbf{F}_q)$. Its complete projective deepest-syndrome locus is

$$U(A) = \mathbf{P}^2(\mathbf{F}_q) \setminus \bigcup_{\{a,b\} \subset A} \overline{ab}.$$

Thus $u \in U(A)$ exactly when adjoining u to a parity-check matrix for A gives a one-column MDS extension.

2 Redundancy three: determinant atlases

Let

$$A = (\mathbf{P}(h_1), \dots, \mathbf{P}(h_6)) \subset \mathbf{P}(V), \quad \dim V = 3,$$

and choose a volume form $\omega \in \det(V)^*$. For $u \in U(A)$ define

$$d_{ij}(u) = \omega(u, h_i, h_j).$$

The following hold in every characteristic, including supports containing the point at infinity.

1. Modulo the exact edge-torus action

$$d_{ij} \mapsto \lambda a_i a_j d_{ij},$$

balanced four-cycle monomials generate the full invariant quotient. They separate finite-field torus orbits without any square-root choice.

2. For rank-two syndromes, and support size at least five, the four-cycle atlas reconstructs the syndrome.
3. The only structural contraction is the rank-one, or conic, locus. On that locus all raw balanced atlases coincide; the unique missing datum is the radical point of the induced alternating form in $\mathbf{P}^1(\mathbf{F}_q) \setminus S$. No higher balanced edge monomial repairs this contraction.
4. Projection from u induces

$$\beta_u(\bar{x}, \bar{y}) = \omega(u, x, y)$$

on V/Fu . The determinant atlas is exactly the Plücker presentation of the labelled projected sextic and hence a point of $M_{0,6}$. There are explicit inverses both from the full Plücker array and from three normalized cross-ratios.

The proof is algebraic invariant theory together with explicit lattice generation for the edge torus and direct Plücker inversion.

2.1 The first sharp rank-one collision

For

$$S = \{0, 1, 2, 3, 4, \infty\} \subset \mathbf{P}^1(\mathbf{F}_{11}),$$

the full semilinear support stabilizer is V_4 , with complement orbits

$$\{5, 10\}, \quad \{6, 7, 8, 9\}.$$

Rank-one syndromes whose radicals lie in these two orbits have the same raw all-one atlas but are sharply separated by the radical stabilizer orbit. Their legal-continuation counts are respectively 11 and 7, and their continuation graphs are

$$K_5 \sqcup C_5 \sqcup K_1, \quad K_5 \sqcup 2K_1.$$

Nontrivial radical stabilizer is the cardinality-minimal intrinsic discriminator. The finite assertions are covered by an atomic certificate and a structurally independent replay.

3 Multi-projection reconstruction and Gale association

Normalize a labelled parent by

$$\begin{aligned} h_1 &= (1, 0, 0), & h_2 &= (0, 1, 0), & h_3 &= (0, 0, 1), \\ h_4 &= (1, 1, 1), & h_5 &= (1, a, b), & h_6 &= (1, c, d). \end{aligned}$$

Each projected sextic supplies one exact linear compatibility row in

$$z = (a, b, c, d, bc, ad).$$

Two coherent projections leave a cubic surface of generic dimension two; three leave a plane cubic of generic dimension one. Four independent projections give a line whose product-compatibility cubic contains the universal collision $h_4 = h_5 = h_6$. Removing that factor leaves a separable quadratic.

Consequently, four abstract coherent projections recover exactly a two-sheeted parent cover, not a rationally unique parent. Explicit examples over $\mathbf{F}_{101}$ and $\mathbf{F}_{256}$, with trivial common diagonal stabilizer, show that the ambiguity persists in odd and even characteristic and after diagonal S_6 -unlabelling.

The deck involution is Gale association. On the normalized chart put

$$L_0 = -\det(h_4, h_5, h_6), \quad L_1 = \det(\nu_2(h_1), \dots, \nu_2(h_6)).$$

The kernel cubic factors as

$$st(L_0s + L_1t).$$

Thus $L_1 = 0$ is exactly the conic locus and is the reduced branch divisor in every characteristic. The equation $L_0 = 0$ is instead an arc-boundary collinearity. Every additional abstract coherent projection preserves the same Gale pair and cannot choose a sheet. The proof uses exact elimination, determinant identities, Gale duality, symbolic certification, and an independent Gale replay.

3.1 Semilinear descent

All gauges, compatibility equations, residual cubics, Gale transforms, branch divisors, and complete-child incidence cuts commute with coefficientwise Frobenius. Off the conic divisor, the only geometric descent obstruction is the C_2 Gale-sheet class:

- a Kummer square class in odd characteristic;
- an Artin–Schreier trace bit in characteristic two.

Connected gauge groups contribute no additional obstruction. If H is the common diagonal stabilizer, an unlabelled sheet descends exactly when the Frobenius sheet bit lies in the image of

$$H \longrightarrow C_2.$$

A surjective image identifies the sheets and sacrifices uniqueness. Over $\mathbf{F}_{q^m}$ the Gale bit is multiplied by m : odd extensions preserve it, and even extensions split it.

The exceptional order-three colour collapse over $\mathbf{F}_8$ is separate: Frobenius acts freely as two 3-cycles on the six colours. This is a lossy C_3 quotient, not a Gale obstruction or a failure of Hilbert 90. The proof combines explicit frame-overlap formulae, finite-field Kummer and Artin–Schreier theory, Hilbert 90 for the connected frame gauges, and a finite-stabilizer image criterion.

4 Complete-child reconstruction

Retaining the literal complete child $L = U(A)$ adds information absent from abstract projections. Parent recovery from selected centres $T \subset L$ is exactly injectivity of the coherent signature map Σ_T , equivalently the intersection of the residual cubic family with the ambient incidence condition

$$U(A') = L.$$

The all-field result is as follows.

- For $q \geq 16$, the complete child alone determines the unlabelled six-arc.
- The same child-rigidity holds for every $q = 13$ fibre.
- In every remaining nonempty recoverable fibre, at most three centres suffice.
- The only failures are empty children and the $q = 7$ two-point conic-complement child. In the latter case all 294 literal parents have the same coherent signature even after both available centres are used.

The complete nontrivial base-size table is

q	$ L $	literal parents	minimum centres
8	4	10	3
9	6	8	3
9	7	2	2
9	8	4	2
9	8	2	2
11	12	22	3

and every other nonempty residual fibre through $q = 13$ is a singleton.

For $q \geq 16$ the proof is a secant-union rigidity argument. For small fields it is an exhaustive classification of child-derived fifteen-line decompositions followed by an exact collision hypergraph. For every pair of parents and every one of the 720 diagonal label transporters, the calculation records the set of centres that distinguish them and computes the exact transversal number. An independent implementation recomputes every emitted parent, atlas, transporter comparison, and hitting set.

4.1 Exceptional non-GRS controls

There are exactly four semilinear classes of non-GRS six-arcs with nonempty deepest-syndrome locus contained in a conic, at

$$q = 8, 9, 9, 11,$$

with child sizes 4, 6, 7, 12 and projective deep-hole orbit profiles

$$4, \quad 6, \quad 1 + 6, \quad 12.$$

A deletion-trace invariant, obtained by intersecting the unique conic through each five-point deletion with the complete child, recovers the parent in the six-point $q = 9$ and twelve-point $q = 11$ cases, but not in the $q = 8$ or seven-point $q = 9$ cases.

For the recovering six-point $q = 9$ fibre, the eight parents form two sheets of four perfect matchings of K_6 minus a perfect matching. Their shared-trace graph is

$$K_{4,4} \setminus 4K_2 \cong Q_3.$$

The order-48 fixed-child group is the full cube automorphism group. This example passes decorated recovery but fails the proposed modular refinement: the relevant Gram rank is 3, and the Sylow- C_3 augmentation module is projective and hence zero in the stable category. The stronger coherent determinant atlas recovers all four exceptional families with exact minimum centre counts

$$3, \quad 3, \quad 2, \quad 3.$$

5 Redundancy five: complete classification of $\text{PRS}(q-4)$

For every prime power $q \geq 7$,

$$\rho(\text{PRS}(q-4)) = 4.$$

This is unconditional, by the Seroussi–Roth completeness range for the quartic normal rational curve.

For a syndrome $f = (a_0 : \cdots : a_4)$, let

$$H_f = \begin{pmatrix} a_0 & a_1 & a_2 & a_3 \\ a_1 & a_2 & a_3 & a_4 \end{pmatrix}.$$

Off the curve, $\ker H_f$ is a pencil of binary cubics. A characteristic-free recurrence and confluent-Vandermonde argument proves

$$f \text{ is deep} \iff \ker H_f \text{ has no totally split squarefree cubic.}$$

The complete PFL_2 classification consists of:

- the tangent family, with $q(q+1)$ points;
- the conjugate σ -secant family, with

$$\frac{q(q+1)(q-1)}{2}$$

points;

- if $q \equiv 2 \pmod{3}$, rational osculating-pair intersections, with $q(q+1)/2$ points;
- if $q \equiv 1 \pmod{3}$, conjugate osculating-pair intersections, with $q(q-1)/2$ points;
- in characteristic 3, the common nucleus

$$(0 : 0 : 1 : 0 : 0)$$

of all osculating planes, together with one wild Artin–Schreier orbit of size $(q^2 - 1)/2$ and stabilizer $2q$, selected by a nonsquare condition.

The arithmetic mechanism is exact. Tame cyclic degree-three covers are deep precisely when their order-three deck transformation is Frobenius-twisted. In characteristic three, the corresponding condition is that the additive map $z \mapsto z^3 + az$ have irrational kernel.

For geometrically S_3 covers, the off-diagonal fibre-square curve has bidegree $(2, 2)$, arithmetic genus one, and is absolutely irreducible. Aubry–Perret gives

$$\#Y(\mathbf{F}_q) \geq q - 2\sqrt{q}.$$

After deleting at most twelve diagonal and branch points, a split cubic exists for every $q \geq 23$. Hence no sporadic orbit survives above 19.

5.1 Complete sporadic table

The sporadic classification is organized by the Frobenius type of the degree-four branch divisor and, in the fully split case, by its j -invariant:

branch type	stabilizer	fields and orbit sizes
$1 + 1 + 1 + 1, j = 0$	A_4	$7 : 28, 13 : 182, 19 : 570$
$1 + 1 + 1 + 1, j \neq 0$	V_4	$9 : 180, 11 : 330, 13 : 546, 17 : 1224$
$1 + 1 + 2$	C_2	$7 : 2 \times 168, 8 : 3 \times 252, 9 : 2 \times 360, 11 : 660$
$2 + 1 + 1$ with a cusp	C_2	$7 : 168$
$1 + 3$	C_3	$7 : 112$

The A_4 cases are exactly equianharmonic branch tetrads, characterized by $j = 0$, equivalently by vanishing of the binary-quartic invariant I . The three size-252 orbits over $\mathbf{F}_8$ form a free $\text{Gal}(\mathbf{F}_8/\mathbf{F}_2)$ -torsor and fuse semilinearly.

These sporadic configurations persist geometrically over larger fields but cease to be deep because split fibres appear. Their geometry is uniform; their deepness is a bounded-field arithmetic phenomenon.

The proof combines characteristic-free Hankel apolarity, confluent Vandermonde arguments, stratification by pencil gcd and degree-three monodromy, finite-field covering-curve estimates, and a two-implementation exhaustive census for every prime power $7 \leq q \leq 49$. A claim-specific literature audit found no pre-emption of the new families or completeness theorem, subject to its stated database-coverage caveat.

6 Redundancy six: all-field existence and orbit counts

For $\text{PRS}(q - 5)$, the covering radius is 5 for every $q \geq 7$. A syndrome is deep exactly when its Hankel-kernel net of binary quartics has no totally split squarefree quartic.

The persistent stratum has a quadratic, rather than growing-degree, common factor and contains exactly

$$\frac{q(q+1)^2}{2}$$

points. It consists of tangent fibres, whose common quadratic is a rational square, and σ -fibres, whose common quadratic is irreducible.

For tangent fibres, if $p \neq 5$ there is one orbit with

$$(|O|, |\text{Stab}|) = (q(q+1), q-1).$$

In characteristic five it splits into

$$(q+1, q(q-1)), \quad (q^2-1, q).$$

For an irreducible quadratic, identify the fibre with the norm-one torus T . Its stabilizer acts by $z \mapsto z^5$ and inversion. If $5 \nmid q+1$ there is one orbit. If $5 \mid q+1$, then $T/T^5 \cong C_5$ modulo inversion gives three orbits with stabilizers 10, 5, 5. Frobenius acts on C_5 by multiplication by p .

The complete trivial-gcd exceptional inventory is

q	7	8	9	11	13
# PGL_2 -orbits	18	11	4	2	1

together with one $q + 1$ point characteristic-two nucleus orbit for every

$$q = 2^m, \quad m \text{ odd}, \quad m \geq 5,$$

and no other trivial-gcd orbits.

The nucleus family is represented by

$$\langle 1, t, t^4 \rangle$$

and is deep exactly when m is odd. One odd-characteristic echo occurs at $q = 11$, where six collision lines exhaust the twelve rational points of the relevant conic. The second $q = 11$ orbit has normal form

$$W = \langle u, tu, u^2 + 5 \rangle, \quad u = t^2 - 2,$$

with all fifteen rational split-quadratic factor candidates sharing a root. The remaining small-field orbit tables are completely certified, although intrinsic normal forms for every orbit at $q = 7, 8, 9, 13$ have not yet been compressed into a final conceptual list.

The large-field proof uses the first-polar line

$$r \mapsto (a_1 - ra_0, \dots, a_5 - ra_4) \subset \mathbf{P}^4$$

to reduce quartic-net splitting to the redundancy-five cubic-pencil classification. Maximal minors of the quintic catalecticant bound intersections with the secant cubic by 3; the tame cyclic locus costs at most 4; pointed g_4^2 ramification costs at most 6. In characteristics three and two, possible contained polar lines are classified separately.

The computational component is an elementwise exhaustive census in eleven fields, with independent definition-based and Hankel-based scans, complete orbit closure, stabilizer and Frobenius checks, and a separately implemented replay.

7 Redundancy seven: interim theorem

Let $f = (a_0 : \dots : a_6)$ be a binary sextic syndrome and

$$H_f = \begin{pmatrix} a_0 & a_1 & a_2 & a_3 & a_4 & a_5 \\ a_1 & a_2 & a_3 & a_4 & a_5 & a_6 \end{pmatrix}.$$

For $r \in \mathbf{P}^1(\mathbf{F}_q)$, contraction gives

$$b(r) = (a_1 - ra_0, \dots, a_6 - ra_5),$$

with the evident value at infinity, and

$$H_f((t - r)g) = H_{b(r)}g.$$

The contractions trace a first-polar line in redundancy-six syndrome space. Thus f is non-deep exactly when some contraction has a totally split squarefree quartic avoiding the marked root r . The avoidance condition is essential.

Let

$$C_f = \begin{pmatrix} a_0 & a_1 & a_2 & a_3 & a_4 \\ a_1 & a_2 & a_3 & a_4 & a_5 \\ a_2 & a_3 & a_4 & a_5 & a_6 \end{pmatrix}.$$

The polar line lies entirely in the redundancy-six catalecticant secant closure if and only if $\text{rank } C_f \leq 2$. After removing rank-one points and rational split secants, this leaves exactly the persistent tangent and conjugate- σ stratum, again with

$$\frac{q(q+1)^2}{2}$$

points.

7.1 Persistent orbit law

The complete orbit law is obtained from T/T^6 modulo inversion and Frobenius. Put $d = \gcd(6, q+1)$. An inversion-fixed class gives

$$(|O|, |\text{Stab}|) = \left(\frac{q(q^2-1)}{2d}, 2d \right),$$

whereas a paired class gives

$$(|O|, |\text{Stab}|) = \left(\frac{q(q^2-1)}{d}, d \right).$$

Frobenius acts by multiplication by p on C_d . On tangent fibres, if $p \nmid 6$ there is one orbit

$$(q(q+1), q-1).$$

In characteristics two and three there are two:

$$(q+1, q(q-1)), \quad (q^2-1, q).$$

These formulae have been independently replayed at $q = 4, 5, 7, 8, 9$, realizing $d = 1, 2, 3, 6$ and both modular cases.

7.2 The central characteristic-two point

The redundancy-six nucleus line contracts to the single central syndrome e_3 . Its web is

$$\langle 1, t, t^4, t^5 \rangle.$$

It is deep exactly over $\mathbf{F}_{2^m}$ with m odd. For even m , the polynomial $t^4 + t$, together with the point at infinity, gives a split quintic on $\mathbf{P}^1(\mathbf{F}_4)$.

7.3 High-field theorem

For every prime power $q \geq 37$,

$$\text{Deep}(\text{PRS}(q-6)) = \{\text{persistent tangent}/\sigma \text{ stratum}\} \cup \begin{cases} \{e_3\}, & q = 2^m, \ m \text{ odd}, \\ \emptyset, & \text{otherwise.} \end{cases}$$

Consequently the number of deep syndrome directions is

$$\frac{q(q+1)^2}{2} + \begin{cases} 1, & q = 2^m, \ m \text{ odd}, \\ 0, & \text{otherwise.} \end{cases}$$

The proof has three quantitative ingredients:

1. a pointed trivial-gcd upgrade of the redundancy-six splitting theorem, using the genus-one fibre estimate with one additional forbidden root, valid for $q \geq 37$;
2. a bidegree union bound for gcd-one contractions, proving the existence of a split lift for $q \geq 16$ unless the common root equals the marked contraction point;
3. identification of that equality locus with ramification of a separable g_5^3 , of degree at most 8.

Together with at most three secant-closure parameters and at most one characteristic-two nucleus parameter, the total exceptional budget is

$$3 + 8 + 1 = 12 < q + 1.$$

What remains open is exactly the orbit-reduced calibration for $q < 37$ and conceptual normal forms for any additional exceptional webs. No complete all-field redundancy-seven classification is asserted.

8 Proof architecture and present boundary

The results use four complementary types of proof.

1. **Exact algebraic proofs:** determinant factorization, torus invariant lattices, Plücker inversion, catalecticant rank criteria, Hankel apolarity, confluent Vandermonde arguments, Gale duality, and first-polar contraction identities.
2. **Arithmetic-geometric proofs:** classification of degree-three cover monodromy, Kummer and Artin–Schreier descent, norm-one torus orbit arithmetic, Aubry–Perret bounds on genus-at-most-one fibre-square curves, ramification-degree bounds, and bidegree union bounds.
3. **Finite group and incidence proofs:** stabilizer-image descent criteria, secant-union rigidity, exact-cover reconstruction, and collision hypergraph transversal calculations.
4. **Certified bounded classifications:** deterministic finite-field censuses, exact orbit closure and stabilizer computations, frozen compact certificates, and independently implemented replays. Where a replay does not independently establish exhaustiveness, that trusted boundary is stated explicitly.

These results do not prove the general Reed–Solomon deep-hole conjecture. What is complete is:

- the all-field redundancy-three reconstruction theory, including its Gale ambiguity, semilinear descent, and exact small-field exceptions;
- the all-field projective-semilinear deep-hole classification at redundancy five;
- the all-field existence and orbit-count classification at redundancy six, with some small exceptional orbits still awaiting conceptual normal forms;
- the redundancy-seven classification for $q \geq 37$, together with the exact persistent orbit law and central characteristic-two exception.